Compton Scattering of Gamma Rays

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Abstract

The Compton effect provides direct experimental evidence for the particle nature of electromagnetic radiation through the scattering of photons by electrons through collision. In this experiment, the energy of gamma rays scattered from a solid target was measured as a function of scattering angle using a NaI(Tl) scintillation detector and multichannel analysis. By exploiting the linearised Compton relation, a linear dependence was observed between the inverse of the scattered photon energy and $1-\cos\theta$. A weighted fit accounting for angular uncertainty and statistical variance yielded a slope $m = (1.988 \pm 0.022)\,\mathrm{MeV}^{-1}$ and intercept $c = (1.576 \pm 0.008)\,\mathrm{MeV}^{-1}$, corresponding to an inferred electron rest energy of $m_e c^2 = (0.503 \pm 0.006)\,\mathrm{MeV}$ and an incident photon energy of $E_\gamma = (0.635 \pm 0.003)\,\mathrm{MeV}$. The inferred electron rest energy differs from the accepted value of 0.511 by 1.6% and is consistent within the experimental uncertainty. The resulting reduced chi-squared of $\chi^2_\nu = 0.74$ indicates a statistically consistent description of the data, with uncertainties that are conservative particularly the angular. In addition, the measured differential cross-section was compared with the Klein–Nishina prediction, showing good agreement over the majority of the angular range, with systematic deviations at small scattering angles attributable to detector resolution, angular misalignment, and additional scattering processes.

Theory

Compton scattering describes the collision of a high-energy photon with a free or weakly bound electron, resulting in a reduction in photon energy and a corresponding increase in wavelength. Classical electromagnetic theory predicts that radiation scattered from electrons should retain its original wavelength; however, this prediction fails at X-ray and gamma-ray energies. Treating the photon as a particle with energy $E_\gamma = h\nu$ and momentum $p_\gamma = h/\lambda$, and applying relativistic conservation of energy and momentum to the photon–electron collision, leads to the Compton scattering formula [1].

For an incident photon of energy E_γ scattered through an angle θ, the energy of the scattered photon E'_γ is

$$E'_\gamma = \frac{E_\gamma}{1 + \frac{E_\gamma}{m_e c^2}(1 - \cos\theta)}, \tag{1}$$

where $m_e c^2 = 0.511\,\text{MeV}$ is the electron rest energy. For the ^{137}Cs source used in this experiment, $E_\gamma = 0.662\,\text{MeV}$, giving $\alpha = E_\gamma/(m_e c^2) \approx 1.29$.

Rearranging yields the linearised form

$$\frac{1}{E'_\gamma} = \frac{1}{E_\gamma} + \frac{1}{m_e c^2}(1 - \cos\theta), \tag{2}$$

which predicts a straight-line relationship between $1/E'_\gamma$ and $1 - \cos\theta$. The slope of this relation corresponds directly to $(m_e c^2)^{-1}$, providing a direct experimental test of relativistic energy–momentum conservation and the particle nature of this radiation.

The angular dependence of the Compton scattering cross-section is given by the Klein–Nishina formula [2],

$$\frac{d\sigma}{d\Omega} = \frac{r_0^2}{2}\left(\frac{E'}{E}\right)^2\left(\frac{E'}{E} + \frac{E}{E'} - \sin^2\theta\right), \tag{3}$$

where r_0 is the classical electron radius, E is the incident photon energy, E' is the scattered photon energy, and θ is the scattering angle. The ratio E'/E is determined by the Compton scattering relation.

Methodology

The experiment was performed using a high-activity ^{137}Cs gamma-ray source mounted in a lead shield with an opening to produce a collimated beam. A cylindrical aluminium scattering rod was positioned at the centre of rotationin front of the source aperture. Scattered photons were detected using a $2'' \times 2''$ NaI(Tl) scintillation detector coupled to a photomultiplier tube, mounted on a rotatable arm which was centred on the scattering target a set distance from the scattering rod.

Energy spectra were recorded using a multichannel analyser operating with MAESTRO-32 software. Prior to data collection, the detector was energy-calibrated using known gamma-ray photopeaks to establish a channel-to-energy relationship. No statistically significant deviation from linearity was observed over the calibrated energy range. Scattering spectra were acquired for angles between 20° and 90° in steps of 5°. For each angle, data were collected until at least 10^4 net counts were obtained under the scattered photon photopeak.

Uncertainty Analysis

The dominant experimental uncertainties arise from the determination of the scattered photon energy and the scattering angle. Repeated measurements at 20°, 40°, and 60° with 10 counts being taken for 20 and 40 and 5 for 60 to ensure a linear uncertainty fit was suitable for these data points. These readings yielded a consistent centroid repeatability of a 0.79 keV, which was adopted as the representative energy uncertainty. This corresponds to a relative uncertainty of approximately 0.12% at 662 keV, increasing to $\sim 0.3\%$ at the lowest scattered energies.

For the linearised relation $y = 1/E'_\gamma$, the propagated uncertainty is

$$\sigma_y = \frac{\sigma_E}{E'^2_\gamma}.$$

An angular uncertainty of $\sigma_\theta = 1^\circ$ was chosen to account for alignment uncertainty and positioning limitations. Since $x = 1 - \cos\theta$, this propagates as

$$\sigma_x = |\sin\theta|\,\sigma_\theta,$$

with σ_θ expressed in radians. At small angles, this contribution becomes dominant: for example, at $\theta = 20^\circ$, $\sigma_x \approx 0.006$, corresponding to a $\sim 6\%$ relative uncertainty in $1-\cos\theta$, whereas at $\theta = 80^\circ$ the same angular uncertainty contributes less than 1%.

To account for uncertainty in both variables, the fit was performed using an effective variance

$$\sigma^2_{y,\text{eff}} = \sigma^2_y + m^2\sigma^2_x,$$

which incorporates angular uncertainty mapped into the dependent variable. The fit was iterated to convergence, yielding a reduced chi-squared of $\chi^2_\nu = 0.74$. The fact that $\chi^2_\nu < 1$ suggests that the reasoned uncertainties were conservative and could possibly have been sharpened. This value also demonstrates no significant unmodelled systematic effects impacting the linearised analysis.

Results

Linearised Compton relation

The weighted fit yields $m = (1.988 \pm 0.022)\,\text{MeV}^{-1}$ and $c = (1.576 \pm 0.008)\,\text{MeV}^{-1}$. These correspond to $m_e c^2 = (0.503 \pm 0.006)\,\text{MeV}$ and $E_\gamma = (0.635 \pm 0.003)\,\text{MeV}$. The fitted slope differs from the theoretical value $(m_e c^2)^{-1} = 1.956\,\text{MeV}^{-1}$ by 1.6%, well within the experimental uncertainty. The inferred incident photon energy is approximately 4.1% below the expected 0.662 MeV value for ^{137}Cs, consistent with the typival impact of finite

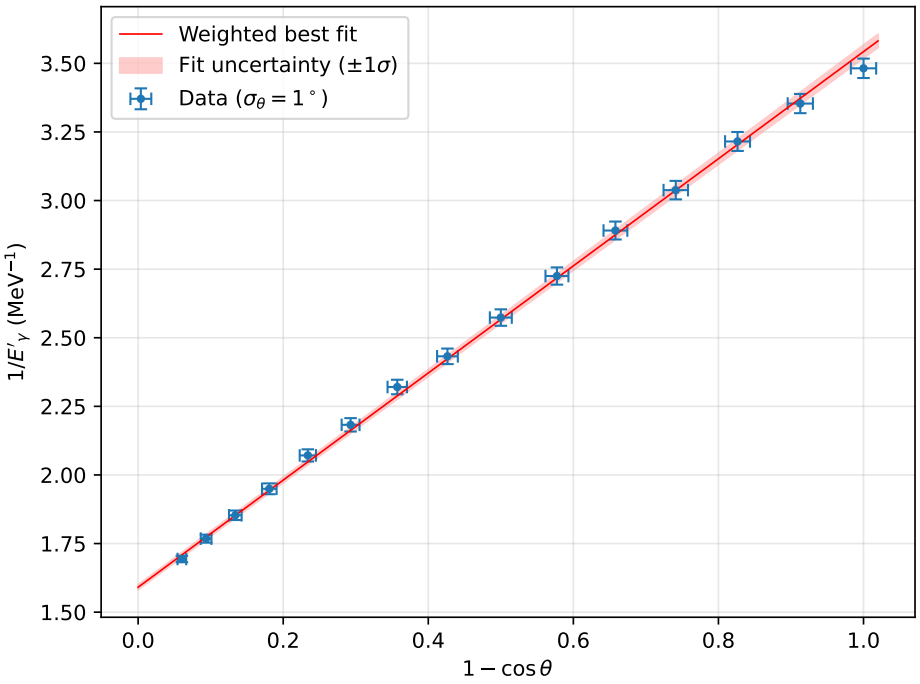


Figure 1: Linearised Compton relation showing $1/E'_\gamma$ plotted against $1-\cos\theta$. The solid line represents a weighted linear fit accounting for angular uncertainty ($\sigma_\theta = 1°$), with the shaded region indicating the $\pm 1\sigma$ confidence band derived from the covariance matrix.

detector resolution and calibration imperfections.

Physical interpretation of the fitted Compton relation

While the linearised representation provides a convenient means of extracting physical parameters, it is instructive to translate the fitted relation back into scattered photon energies in order to assess the physical scale of the results directly. Using the fitted parameters m and c, the scattered energy may be recovered via

$$E'_\gamma(\theta) = \frac{1}{c + m(1-\cos\theta)}.$$

For a representative forward-angle case, $\theta = 20°$, $1-\cos\theta \approx 0.060$ gives

$$E'_\gamma \approx \frac{1}{1.576 + 1.988 \times 0.060} \approx 0.589\,\text{MeV},$$

which is close to the unscattered photopeak energy and consistent with the measured centroid at small angles.

At an intermediate angle, $\theta = 60°$, where $1-\cos\theta = 0.5$, the fit predicts

$$E'_\gamma \approx \frac{1}{1.576 + 0.994} \approx 0.389\,\text{MeV}.$$

This value corresponds well with the mean values of the 5 runs of 60 degrees which were 0.387 MeV and illustrates the substantial energy loss associated with large-angle scattering.

For backscattering at $\theta = 90°$, corresponding to $1 - \cos\theta = 1$, the predicted energy is

$$E'_\gamma \approx \frac{1}{1.576 + 1.988} \approx 0.281\,\text{MeV},$$

in good agreement with the lowest-energy scattered peaks observed experimentally.

Across the full angular range, the recovered energies span approximately 0.59 MeV to 0.28 MeV, consistent with expectations for Compton scattering of 662 keV photons. Residual differences between predicted and measured centroid energies are of order a few keV and are comparable to the intrinsic energy resolution of the NaI(Tl) detector, confirming that the linearised fit provides an accurate physical description of the scattering process.

Differential cross-section

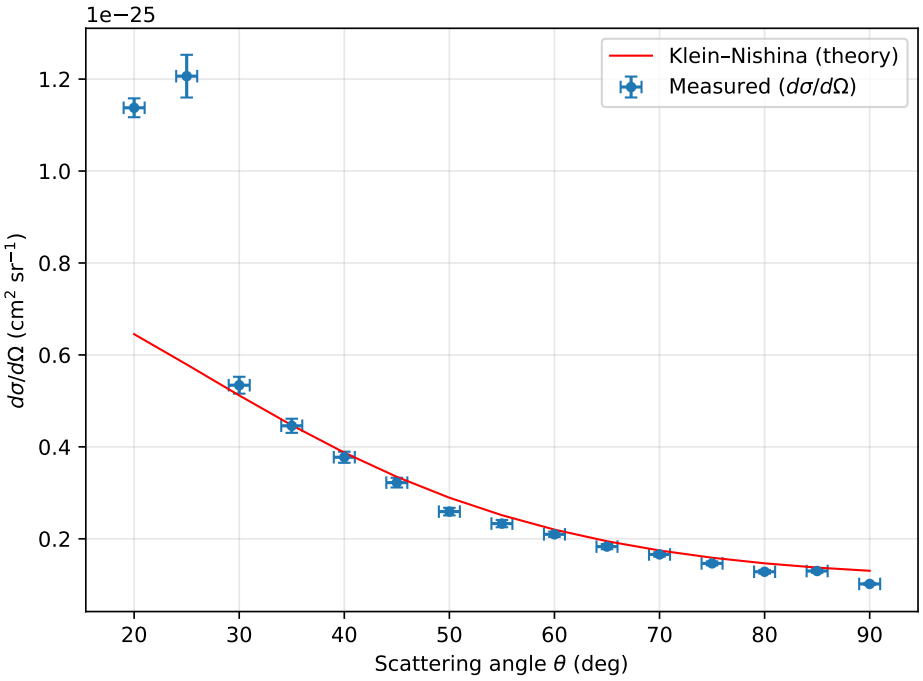


Figure 2: Measured differential cross-section as a function of scattering angle compared with the Klein-Nishina prediction. Error bars represent propagated statistical and angular uncertainties.

The measured differential cross-section follows the Klein-Nishina prediction closely for scattering angles above approximately 30°, where deviations remain within statistical uncertainty. At smaller angles (20°–25°), a systematic excess of order 20–30% is observed relative to the theoretical curve.

Discussion

The systematic deviation observed at the smallest scattering angles arises from several well-understood experimental effects. At low θ, the energy separation between the unscattered photopeak and the Compton-scattered peak becomes comparable to the intrinsic energy resolution of the NaI(Tl) detector (approximately 7% at 662 keV), increasing sensitivity to peak overlap and background subtraction errors [4]. In addition, the strong angular dependence of $1-\cos\theta$ at small θ means that a 1° misalignment can produce changes comparable to statistical uncertainties, disproportionately affecting forward-angle data. Finally, coherent (Rayleigh) scattering and multiple scattering within the aluminium target contribute most strongly at small angles and are not included in the Klein-Nishina formula, which assumes single scattering from free, stationary electrons [3].

Outside this forward angle regime, the measured data show close agreement with the Klein-Nishina prediction. The value of the electron rest energy extracted from the linearised Compton analysis is consistent with the accepted value within experimental uncertainty, providing a quantitative confirmation of relativistic energy - momentum conservation. Similarly, the angular dependence of the measured differential cross-section follows the expected relativistic scattering behaviour over most of the accessible angular range, indicating that deviations are dominated by identifiable experimental limitations rather than a failure of the underlying theory.

Several straightforward modifications could improve the precision of the present measurement. Reducing the angular uncertainty through finer mechanical alignment, as the angluar uncertainty dominated. A lower value then 1 degree might have been a better fit as pointed out by the 0.74 chi-squared value a lower value suggests 1 degree was too conservative. Ideally 10 readings of each angle would be taken resetting the angle each time to mitigate scattering and angular uncertainty. Finally, employing a thinner scattering target would mitigate multiple-scattering contributions, allowing a more direct comparison with the single scattering Klein-Nishina prediction and potentially would scale down values below 30 degrees.

Conclusion

Compton scattering of gamma rays was investigated through measurements of scattered photon energy and angular distribution using a NaI(Tl) scintillation detector. The linearised Compton relation was verified experimentally, yielding parameters consistent with the electron rest energy and the incident photon energy of a ^{137}Cs source within experimental uncertainty. The measured differential cross-section agrees with the Klein-Nishina prediction over most of the angular range, with deviations at small angles attributable to detector resolution, angular alignment, and additional scattering processes. These re-

sults provide a clear experimental demonstration of photon momentum and the quantum nature of electromagnetic radiation.

References

[1] A. H. Compton, "A Quantum Theory of the Scattering of X-rays by Light Elements," *Phys. Rev.* **21**, 483 (1923).

[2] O. Klein and Y. Nishina, "Über die Streuung von Strahlung durch freie Elektronen," *Z. Phys.* **52**, 853 (1929).

[3] W. Heitler, *The Quantum Theory of Radiation*, Oxford University Press (1954).

[4] G. F. Knoll, *Radiation Detection and Measurement*, 4th ed., Wiley (2010).

Appendix A: Python Analysis Code (PDF)

Part 1 Plot

Sigma used to denote uncertainties throughout appendix

In [84]:
```
import numpy as np
import matplotlib.pyplot as plt

#Uncertainty Analysis block

def summary_stats(arr, units=""):
    N = len(arr)
    mean = np.mean(arr)
    std = np.std(arr, ddof=1)
    stderr = std / np.sqrt(N)
    if units:
        print(f"N = {N}, mean = {mean:.5g} {units}, std = {std:.5g} {units}")
    return N, mean, std, stderr
#20 deg
E20_MeV = np.array([0.59926, 0.59698, 0.59795, 0.59786, 0.59833,0.59690, 0.59870

print("20° centroid repeatability")
n20, mean20_MeV, s20_MeV, u20_MeV = summary_stats(E20_MeV, units="MeV")

#40 deg
E40_MeV = np.array([0.48228,  # starred: taken at a slightly different angle fro
    0.48947, 0.48994, 0.48966, 0.48823, 0.49041, 0.48927, 0.48765, 0.48886, 0.49
E40_clean_MeV = E40_MeV[1:]

print("40° centroid repeatability (excluding outlier)")
n40, mean40_MeV, s40_MeV, u40_MeV = summary_stats(E40_clean_MeV, units="MeV")

# 60°
E60_MeV = np.array([0.38857,  # starred: taken at a slightly different angle fro
    0.38731, 0.38721, 0.38621, 0.38792, 0.38619])
E60_clean_MeV = E60_MeV[1:]

print(" 60° centroid repeatability (excluding outlier) ")
n60, mean60_MeV, s60_MeV, u60_MeV = summary_stats(E60_clean_MeV, units="MeV")
```

```
20° centroid repeatability
N = 10, mean = 0.59813 MeV, std = 0.00075286 MeV
40° centroid repeatability (excluding outlier)
N = 10, mean = 0.48933 MeV, std = 0.00086236 MeV
 60° centroid repeatability (excluding outlier)
N = 5, mean = 0.38697 MeV, std = 0.00075194 MeV
```

In [83]:
```
# Since Uncertainties are very similar mean of them was used
sigma_E = np.mean([s20_MeV, s40_MeV, s60_MeV])

print("Adopted energy uncertainty")
print(f"σ_E = {sigma_E:.2g} MeV (centroid repeatability)")
```

```
Adopted energy uncertainty
σ_E = 0.00079 MeV (centroid repeatability)
```

In [86]:
```
# scattering angles (degrees)
theta_deg = np.array([20, 25, 30, 35, 40, 45, 50, 55, 60, 65, 70, 75, 80, 85, 90

# measured scattered photon energies (keV)
E_keV = np.array([590.4, 565.64, 539.66, 512.95, 482.82, 458.16, 430.94, 411.14,

# convert to MeV and form y = 1 / E'
E_MeV = E_keV / 1000.0
y = 1.0 / E_MeV

# representative uncertainty on scattered energy from above block (MeV)
sigma_E = 0.00079
sigma_y = sigma_E / E_MeV**2

# angles in radians
theta = np.deg2rad(theta_deg)

# angular uncertainty
sigma_theta = np.deg2rad(1.0)

# x = 1 - cos(theta) and its uncertainty
x = 1.0 - np.cos(theta)
sigma_x = np.abs(np.sin(theta)) * sigma_theta

# weighted fit y = m x + c
def weighted_line_fit(x, y, sigma):
    w = 1.0 / sigma**2
    S   = np.sum(w)
    Sx  = np.sum(w * x)
    Sy  = np.sum(w * y)
    Sxx = np.sum(w * x * x)
    Sxy = np.sum(w * x * y)

    Delta = S * Sxx - Sx**2
    m = (S * Sxy - Sx * Sy) / Delta
    c = (Sxx * Sy - Sx * Sxy) / Delta

    sigma_m = np.sqrt(S / Delta)
    sigma_c = np.sqrt(Sxx / Delta)

    return m, c, sigma_m, sigma_c

# include x-uncertainty via effective y-uncertainty
sigma_y_eff = sigma_y.copy()

for _ in range(3):
    m, c, sigma_m, sigma_c = weighted_line_fit(x, y, sigma_y_eff)
    sigma_y_eff = np.sqrt(sigma_y**2 + (m * sigma_x)**2)

# goodness of fit
y_fit = m * x + c
chi2 = np.sum(((y - y_fit) / sigma_y_eff)**2)
ndof = len(y) - 2
chi2_red = chi2 / ndof

print("Fit including sigma_theta = 1°:")
print(f"m = {m:.7g} ± {sigma_m:.2g}  (MeV^-1)")
print(f"c = {c:.7g} ± {sigma_c:.2g}  (MeV^-1)")
print(f"chi2_red = {chi2_red:.3f}")
```

```
# inferred physical quantities
me_c2 = 1.0 / m
sigma_me_c2 = sigma_m / m**2
E_gamma = 1.0 / c
sigma_E_gamma = sigma_c / c**2

print(f"m_e c^2 = {me_c2:.6g} ± {sigma_me_c2:.2g} MeV")
print(f"E_gamma = {E_gamma:.6g} ± {sigma_E_gamma:.2g} MeV")


x_line = np.linspace(0, x.max() * 1.02, 400)
y_line = m * x_line + c

# uncertainty band from fit parameters
sigma_line = np.sqrt((x_line * sigma_m)**2 + sigma_c**2)

# plot
fig, ax = plt.subplots()

ax.errorbar(
    x, y, xerr=sigma_x, yerr=sigma_y_eff,fmt="o",markersize=3,capsize=3,elinewid

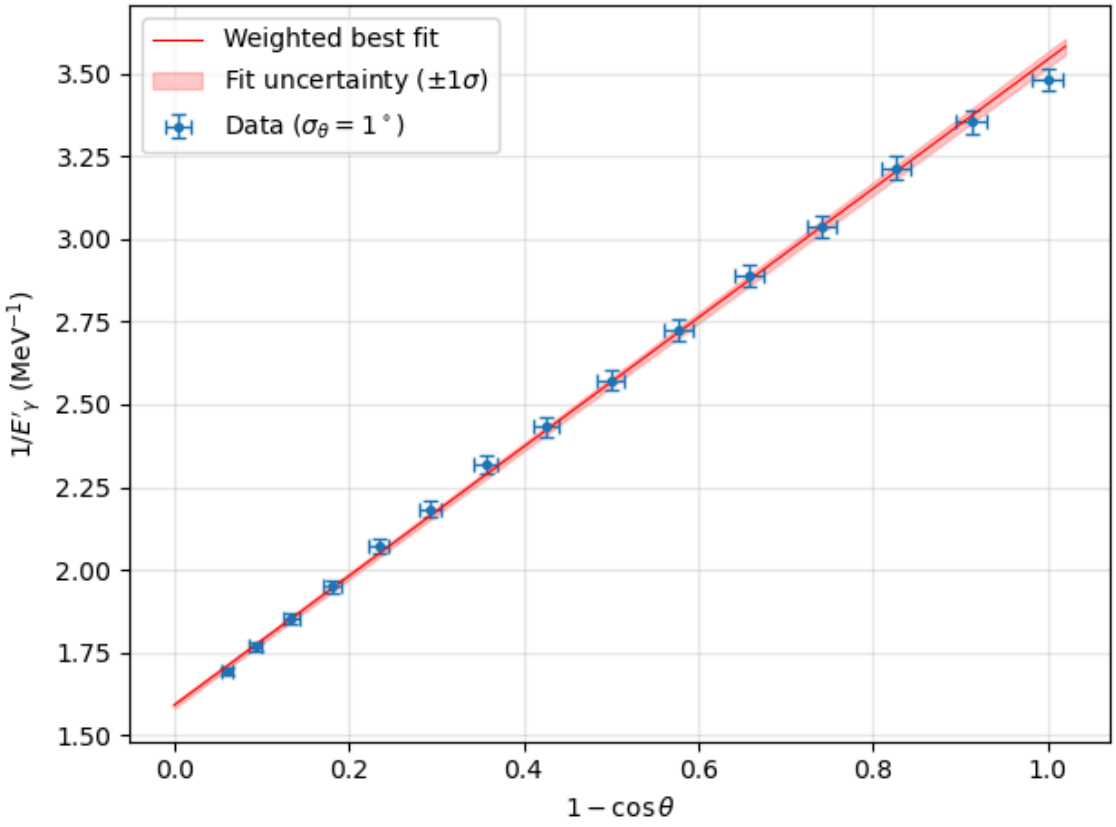
ax.plot(x_line, y_line, color="red", linewidth=0.8, label="Weighted best fit")

ax.fill_between(x_line,y_line - sigma_line, y_line + sigma_line,color="red",alph

ax.set_xlabel(r"$1-\cos\theta$")
ax.set_ylabel(r"$1/E'_\gamma\ \mathrm{(MeV^{-1})}$")
ax.grid(True, alpha=0.3)
ax.legend()

plt.tight_layout()
plt.savefig("Part1.pdf")
plt.show()
```

```
Fit including sigma_theta = 1°:
m = 1.951765 ± 0.022  (MeV^-1)
c = 1.59086 ± 0.0083  (MeV^-1)
chi2_red = 0.686
m_e c^2 = 0.512357 ± 0.0057 MeV
E_gamma = 0.628591 ± 0.0033 MeV
```



Part 2 Code

In [55]:
```
#Theta deg and E Mev used from above

# Net photopeak counts and their uncertainties (after background subtraction)
net_counts = np.array([27485, 10985, 10596, 10578, 10360, 10560, 10504, 10606,
                       10796, 10621, 11002, 10578, 10650, 10488, 11596], dtype=f

net_unc = np.array([496, 423, 364, 365, 332, 345, 336, 361,
                    302, 330, 322, 317, 336, 314, 277], dtype=float)

# Live time per spectrum
time_s = np.array([412.86, 148.23, 306.20, 345.74, 373.52, 420.26, 484.84, 516.1
                   547.66, 578.66, 618.42, 636.70, 686.30, 636.32, 860.34], dtyp

# Photopeak efficiency curve
eff = 0.1522 * (E_MeV ** -1.1325)

# Incident flux at target decay corrected using late November 2025
t_years = 48.3
flux_I = 1.013e6 * np.exp(-t_years / 43.48)   # s^-1 cm^-2

# Number of electrons in aluminium rod
mass_g = 79.3
Z_Al = 13
A_Al_gmol = 26.98
N_A = 6.02214076e23

N_e = (mass_g / A_Al_gmol) * N_A * Z_Al

# Count rate under scattered photopeak
```

```
rate = net_counts / time_s
rate_unc = net_unc / time_s

# Correct for photopeak efficiency
rate_corr = rate / eff
rate_corr_unc = rate_unc / eff

# Per-electron proxy for dσ/dΩ
dsdo_meas = rate_corr / (N_e * flux_I)
dsdo_meas_unc = rate_corr_unc / (N_e * flux_I)

# Klein-Nishina theory

E0 = 0.662       #MeV (Cs-137 gamma energy)
mec2 = 0.511     #MeV
r0_cm = 2.8179403262e-13  # electron radius in cm

th = np.deg2rad(theta_deg)
alpha = E0 / mec2

# Energy ratio E'/E for Compton scattering
Er = 1.0 / (1.0 + alpha * (1.0 - np.cos(th)))

# Klein-Nishina differential cross-section per electron (cm^2 sr^-1)
dsdo_kn = 0.5 * (r0_cm**2) * (Er**2) * (Er + 1.0/Er - np.sin(th)**2)


#Scale factor to compare theory to measured,
#Absolute normalisation is not reliable without a full geometric/attenuation mod
#so one constant scale factor is applied for shape-only comparison.

C = np.mean(dsdo_kn / dsdo_meas)

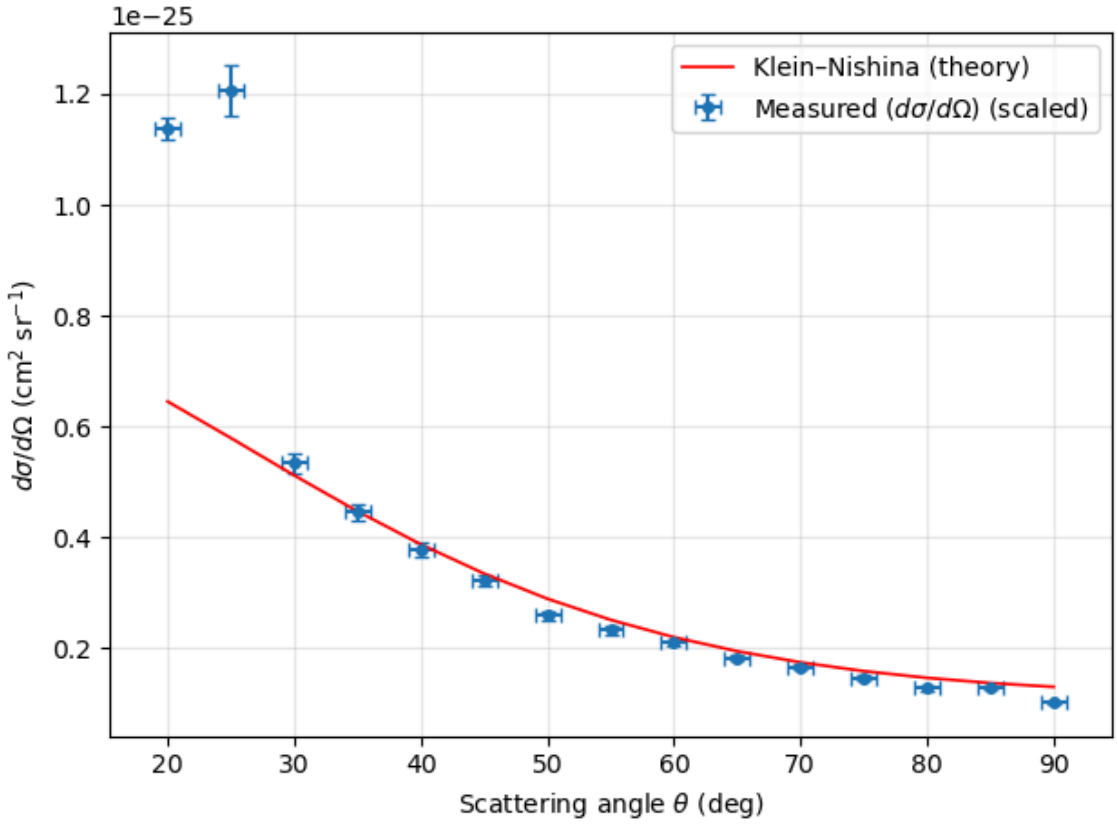
dsdo_meas_scaled = dsdo_meas * C
dsdo_meas_unc_scaled = dsdo_meas_unc * C

# Plot
plt.errorbar(theta_deg,
             dsdo_meas_scaled,
             yerr=dsdo_meas_unc_scaled,
             xerr=1,fmt="o",
             markersize=4,capsize=3,
             label=r"Measured $(d\sigma/d\Omega)$ (scaled)")

plt.plot(theta_deg, dsdo_kn,
    color="red", linewidth=1.2,
    label="Klein-Nishina (theory)")

plt.xlabel(r"Scattering angle $\theta$ (deg)")
plt.ylabel(r"$d\sigma/d\Omega$ (cm$^2$ sr$^{-1}$)")
plt.grid(True, alpha=0.3)
plt.legend()
plt.tight_layout()
plt.savefig("part2.pdf")
plt.show()

print(f"Scale factor C = {C:.3e}")
print(f"Mean efficiency = {np.mean(eff):.3e}")
print(f"N_e (electrons in rod) = {N_e:.3e}")
print(f"Flux I = {flux_I:.3e} s^-1 cm^-2")
```




```
Scale factor C = 3.626e+03
Mean efficiency = 4.319e-01
N_e (electrons in rod) = 2.301e+25
Flux I = 3.336e+05 s^-1 cm^-2
```